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Orbital Data Centers in a Dawn-Dusk SSO

Asael Acosta, Avaneesh Pal

Overview

- Introduction
- Idealized Forces, Full Forces, and Impulsive Maneuvers
  - Methodology
  - Results
- Discussion
- Summary and Future Work

Mission Motivation

- Continuous power requirement for orbital data centers:
    - Dawn–dusk Sun–Synchronous Orbits (SSOs)
- Formation flying enables distributed space infrastructure:
- Earth's oblateness (J_2) dominates long-term orbital dynamics in LEO:
- Accurate J_2 modeling is essential for formation stability:
- Designing relative orbits that exploit J_2 improves mission longevity:

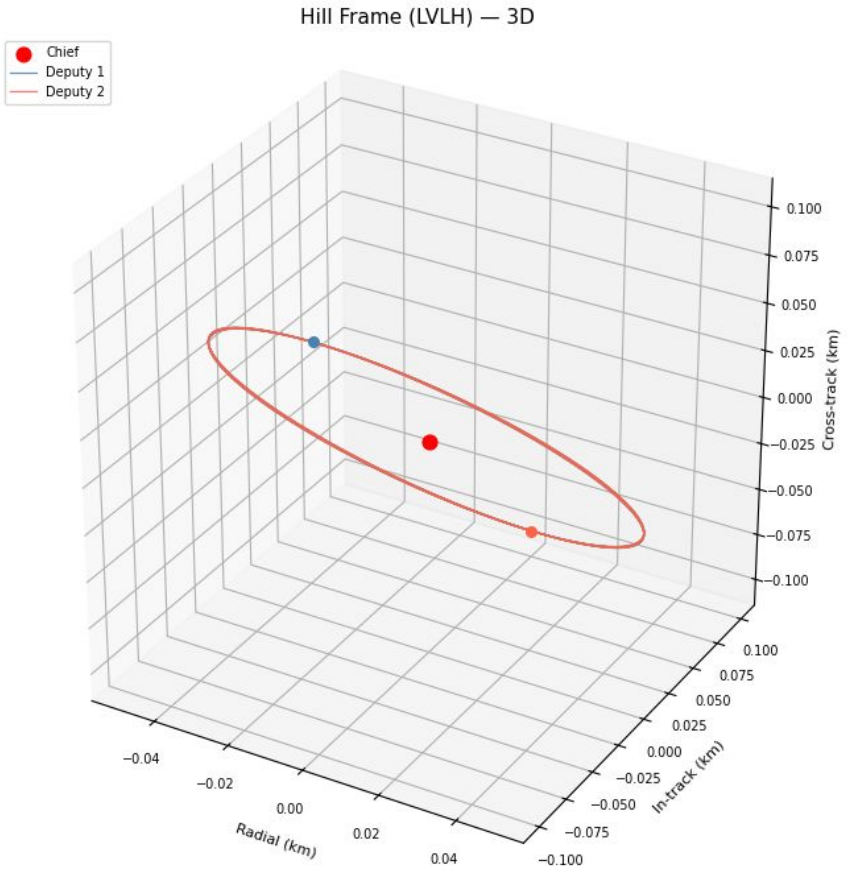
Problem Statement: Design closed or quasi-closed relative orbits for a three-satellite formation in a dawn–dusk SSO using relative orbital elements and mean-element theory.

Assumptions and Simulation Parameters

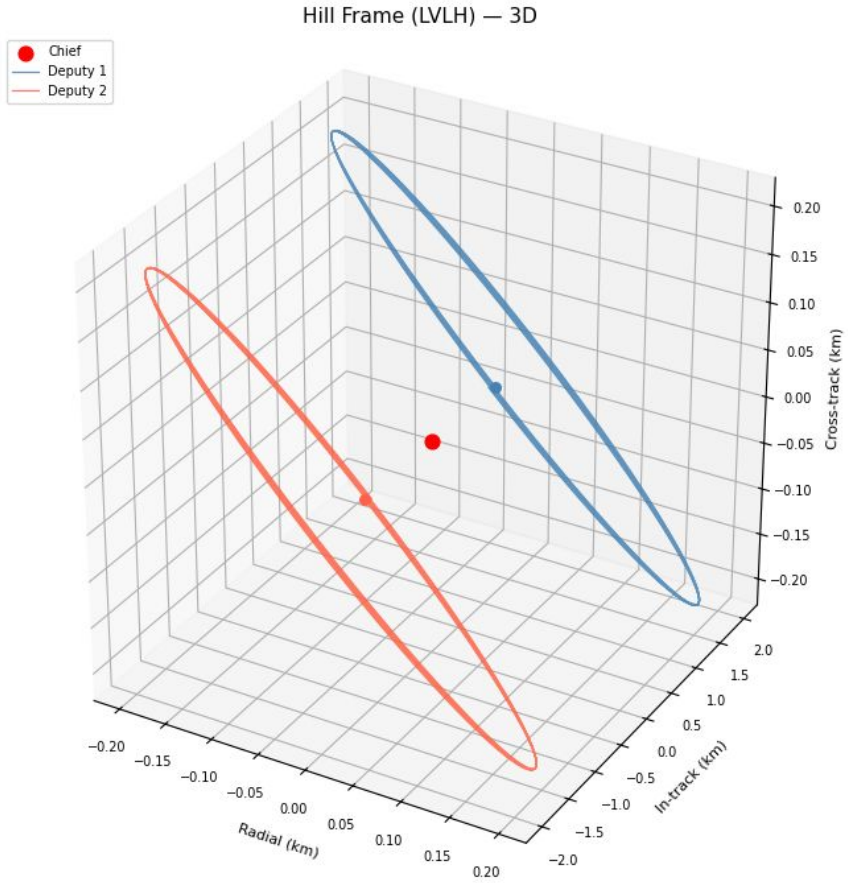
- 3 satellites (1 chief, 2 deputies)
- Cubesat class craft, modeled after ISARA mission
  - Solar Panel Size, Cross Sectional Area, Mass
- ConOps of orbit establishment handled
  - e.g., Launch, Sensor Suite, Choice of Propulsion System
- Altitude of 1000 km selected
- Propagated over 6 months

	Chief-Deputy Distance (km)	Purpose
Design A	0.5 - 1.5 km	Servicing (e.g,, power sharing)
Design B	2 - 5 km	Minimizing station-keeping

Relative Orbit Choices



Design A - Helix



Design B - String of Pearls

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Project Goals

- Idealized Force Model
  - Design Chief - Deputy Model with J_2 Perturbations
  - Minimize Secular Drift
  - Is long-term quasi-periodic relative motion achievable?
- Full Force Model
  - Extend Idealized Force Model to include higher-fidelity gravity model
  - Include Solar Radiation Pressure and Atmospheric Drag
  - Investigate and identify deviations and their causes
- Impulsive Formation Maintenance
  - Design maneuvers to maintain formation within given limits
  - Compare periodic vs. state-triggered maneuver strategies
  - Estimate total ΔV per satellite over 1 year
  - Describe the formation-flying concept of operations.

Work Distribution

- Idealized Force Modeling: Avaneesh Pal
- Full Force Modeling: Avaneesh Pal and Asael Acosta
- Impulsive Maneuvers: Asael Acosta

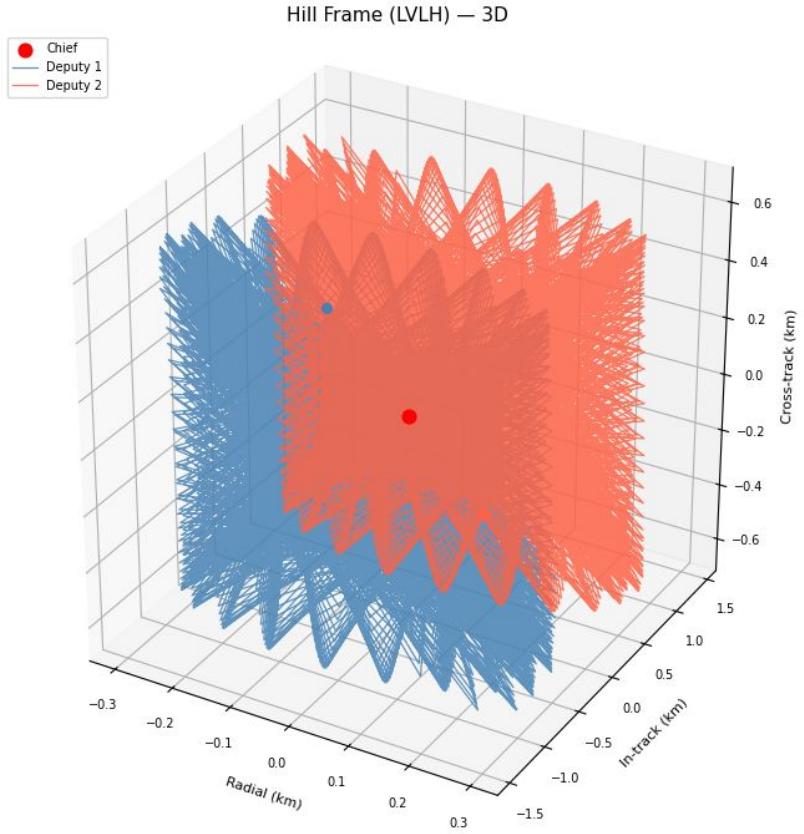
Idealized Force Model

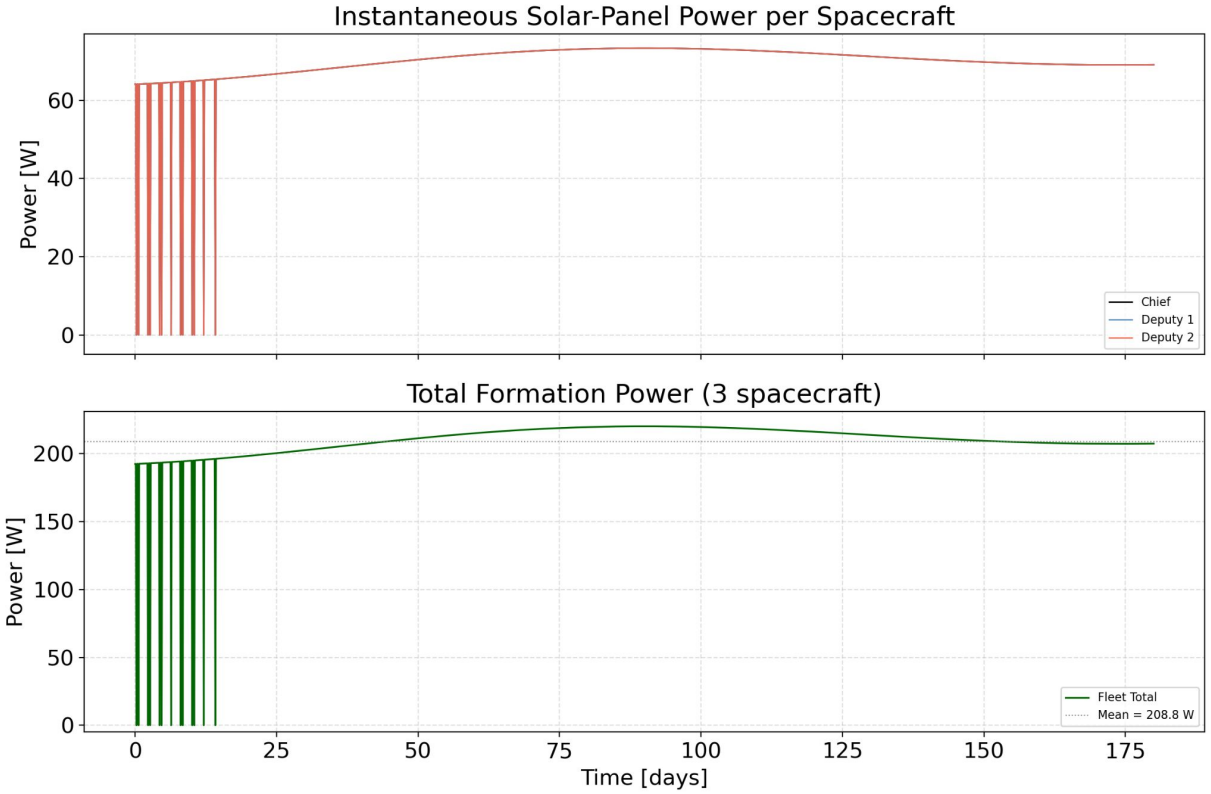
Methodology

- Design closed relative Keplerian Orbits
  - Define Chief Orbit
  - Use HCW equations to find mean motion of chief and use them to define deputy orbits
- Introduce J_2 Perturbations
  - Compute relative motion over time and identify effects on orbital elements
  - Visualize/plot drift
- Modify orbital elements to minimize secular drift
  - Apply J2-invariant relative orbit conditions
  - Assess Stability
- Investigate long-term quasi-periodic motion for all satellites
  - Extend drift-minimizing conditions to deputies
  - Analyze Stability
  - Assess feasibility

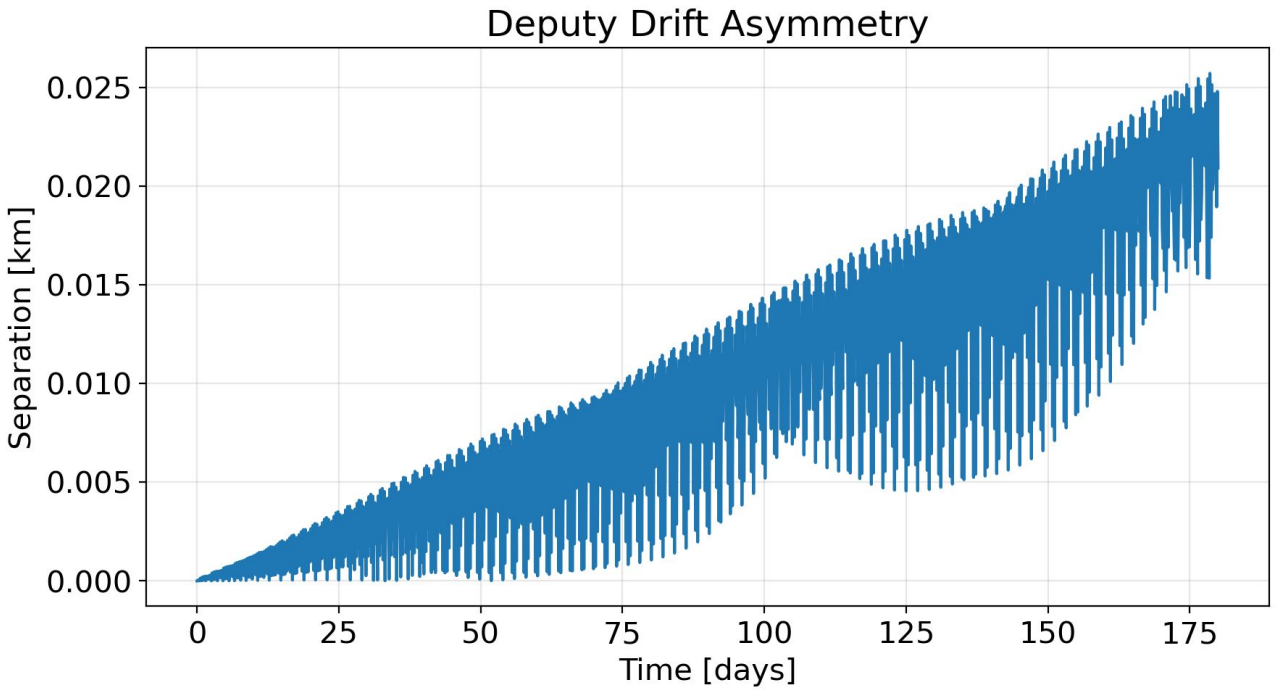
Idealized Force Model Results

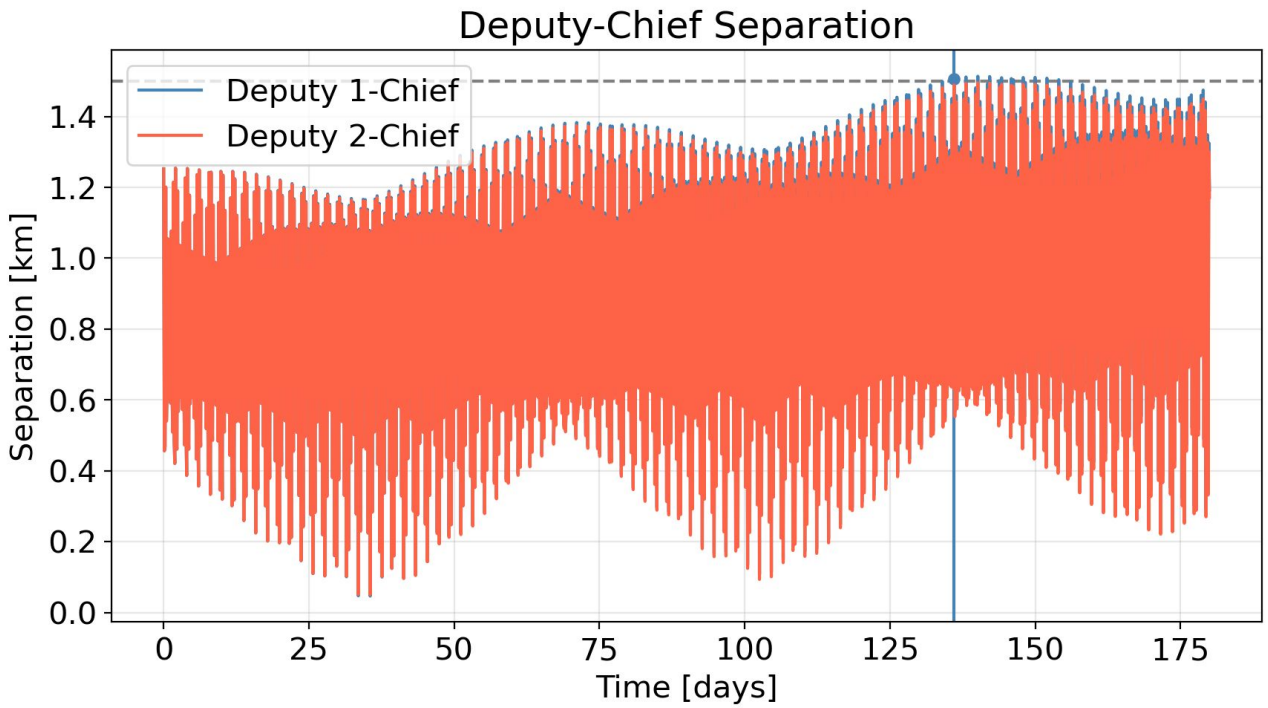
Design A - Trajectory and Solar Power Input





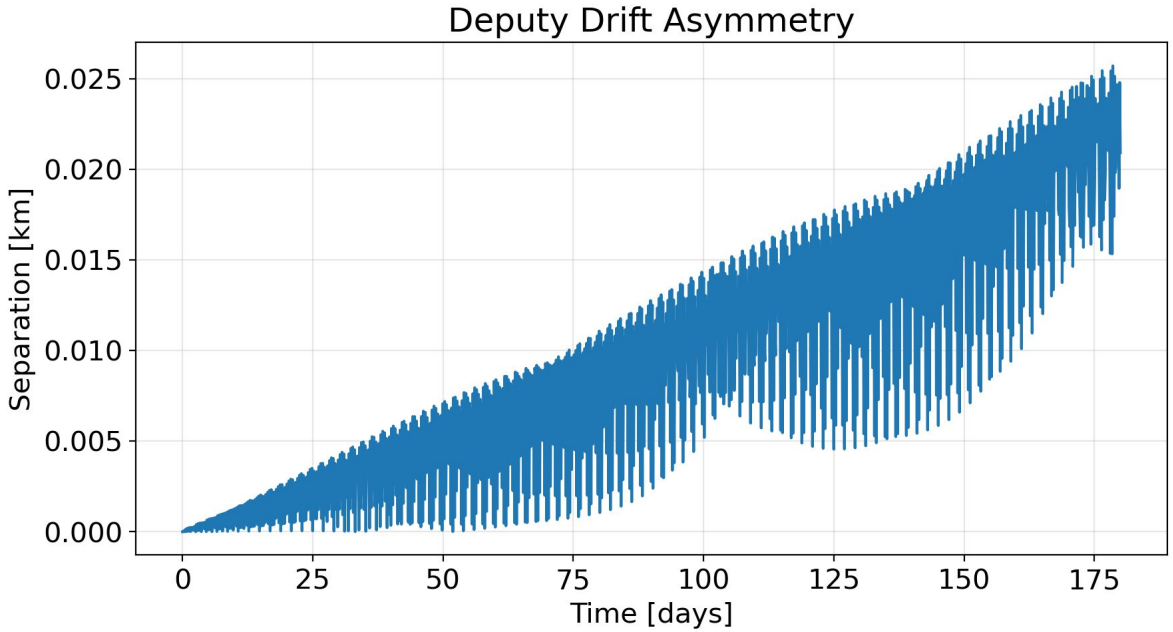
Design A - Asymmetries and Separation



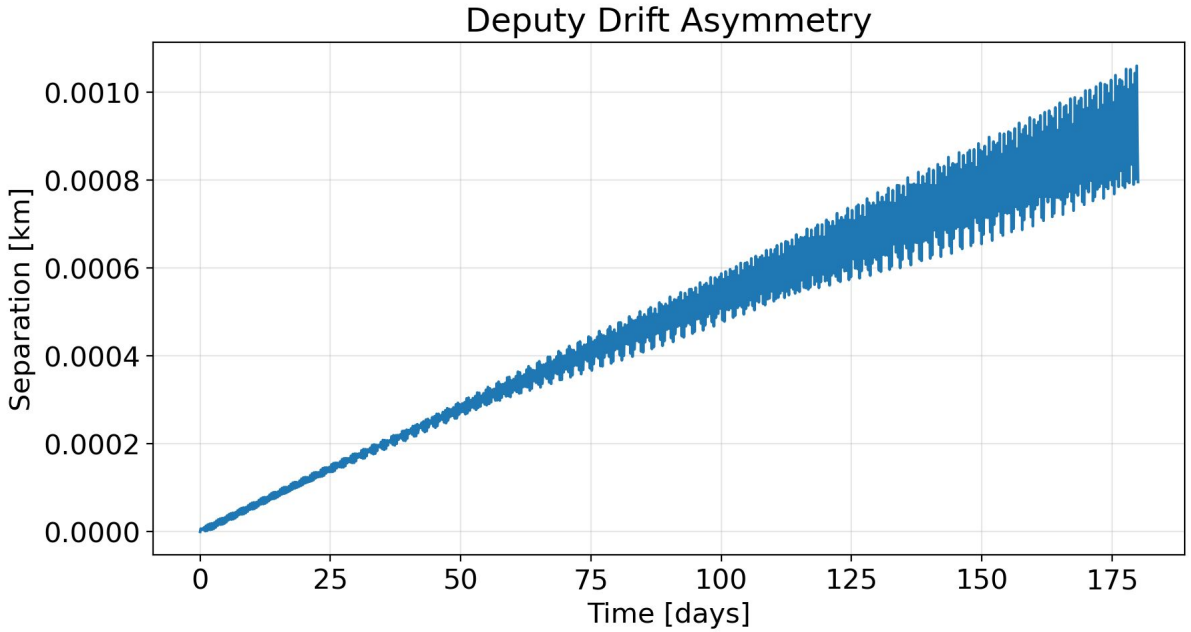


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Design A – Helix vs String of Pearls

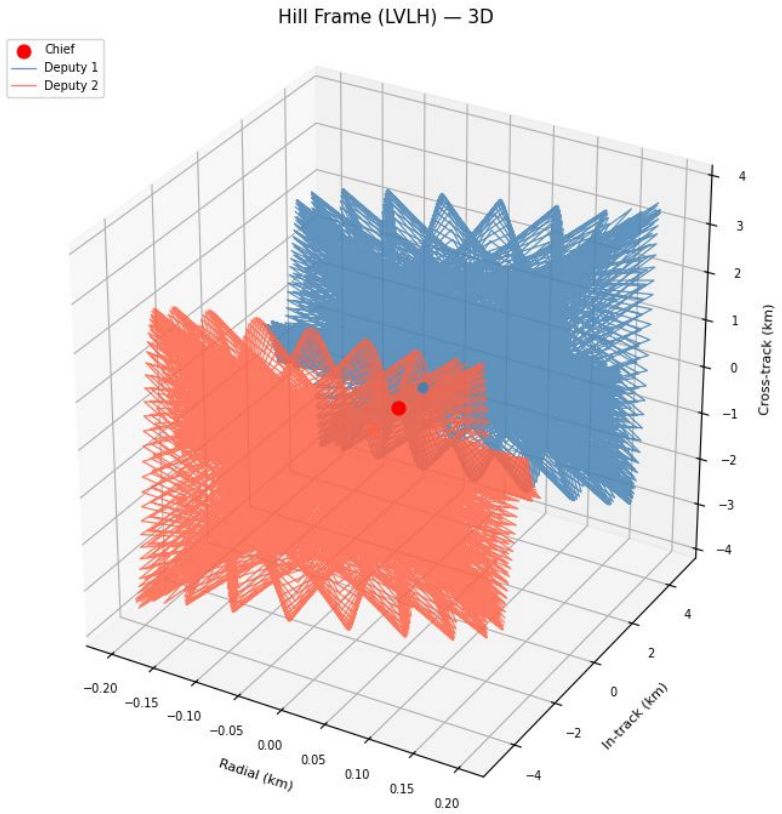


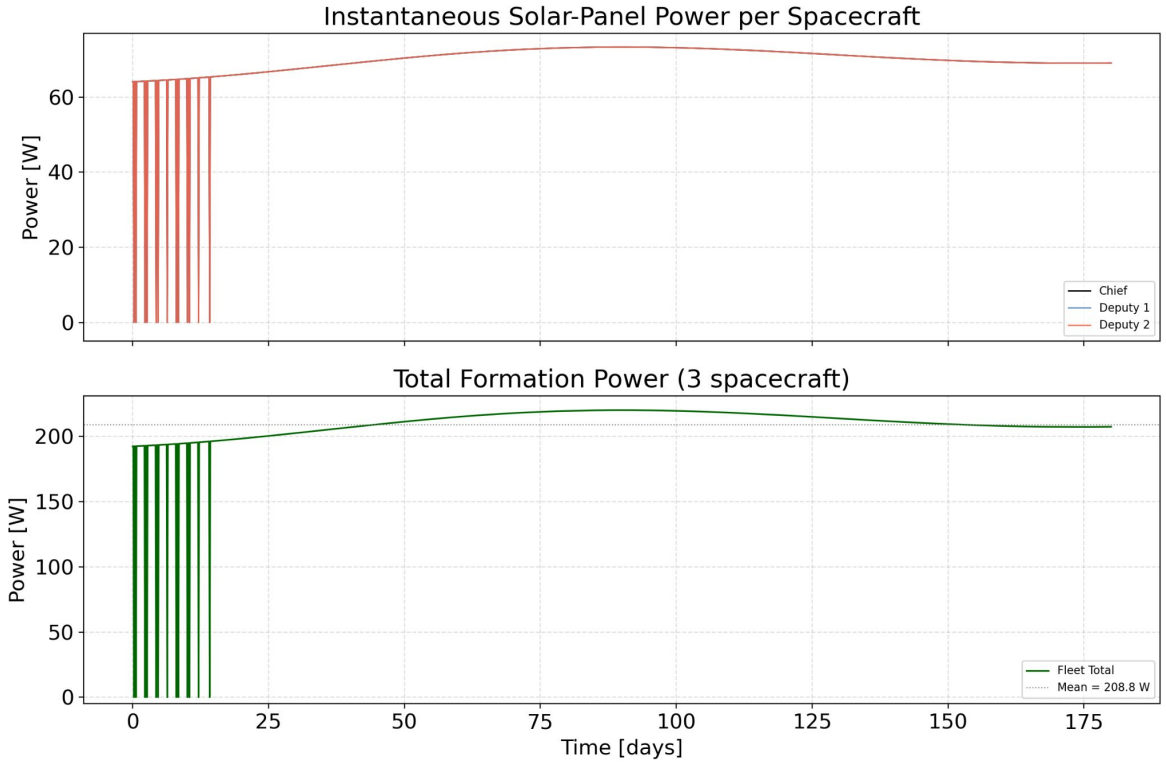
String of Pearls



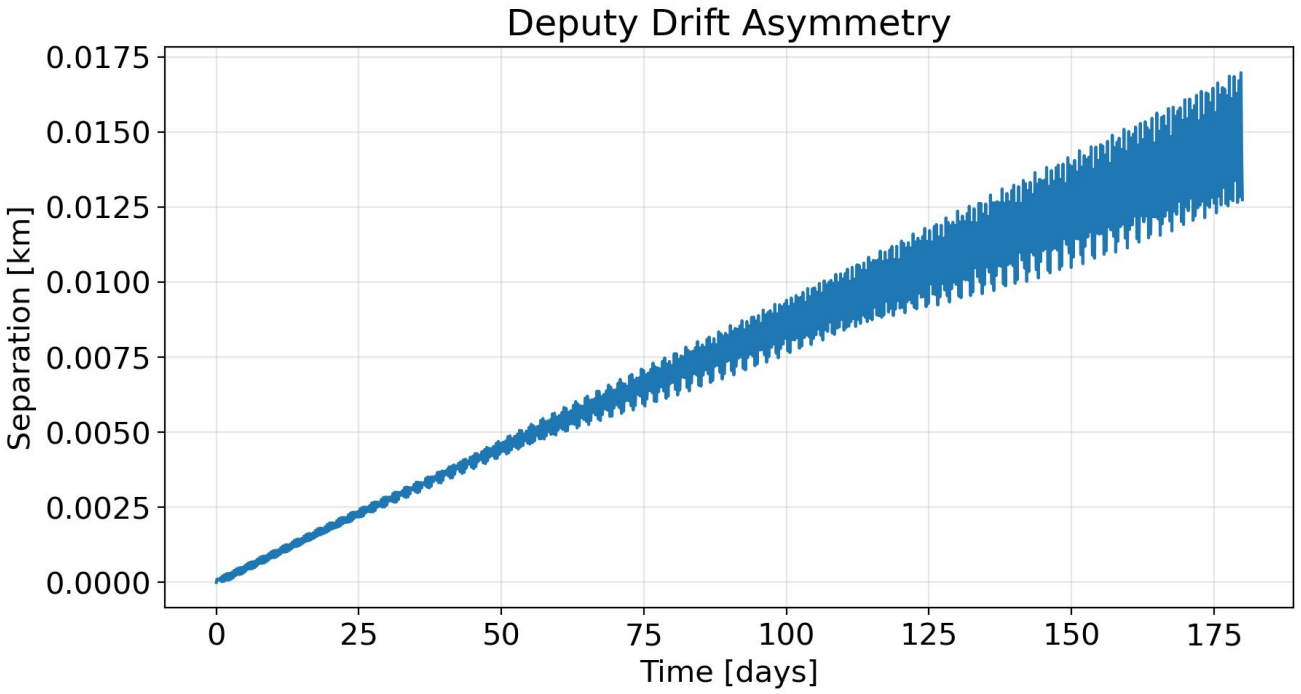
Helix Formation

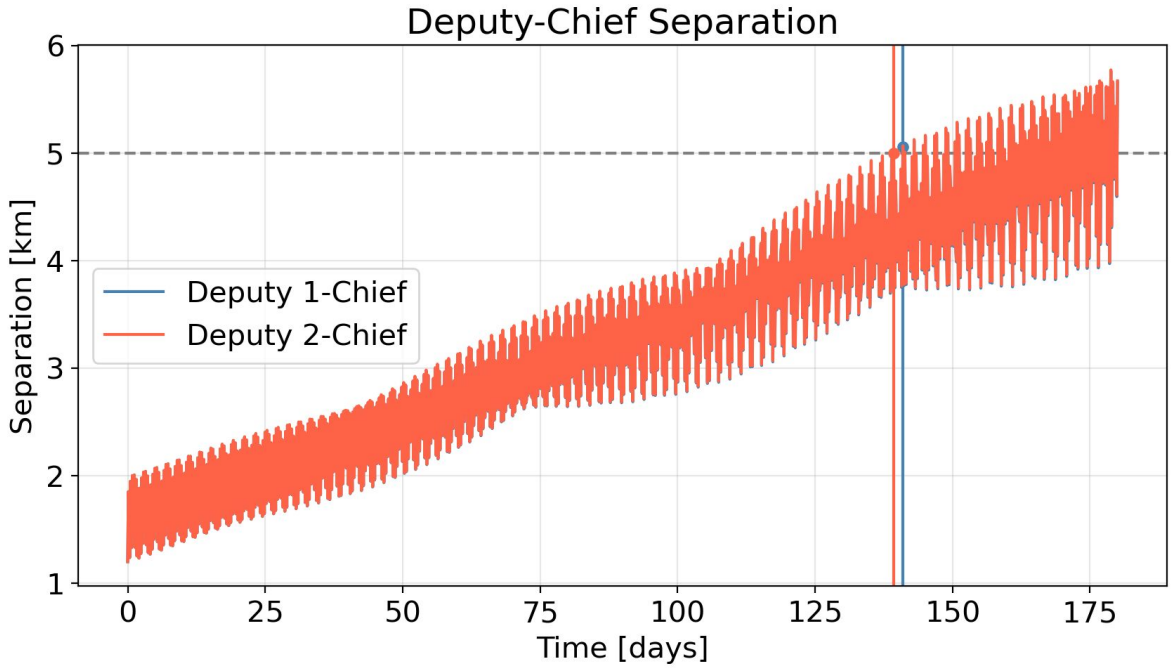
Design B - Trajectory and Solar Power Input





Design B - Asymmetries and Separation





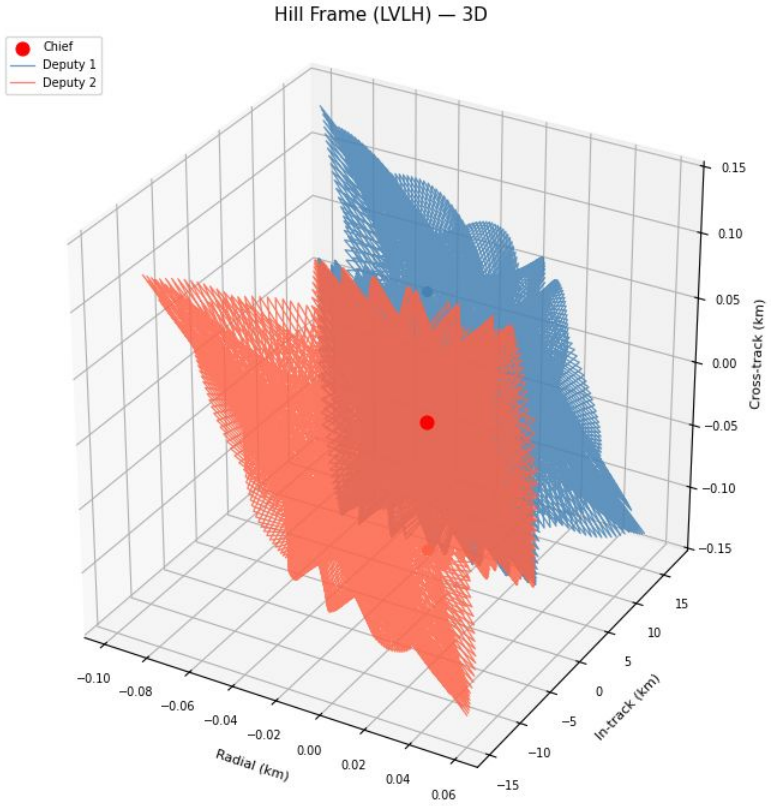
Full Force Model

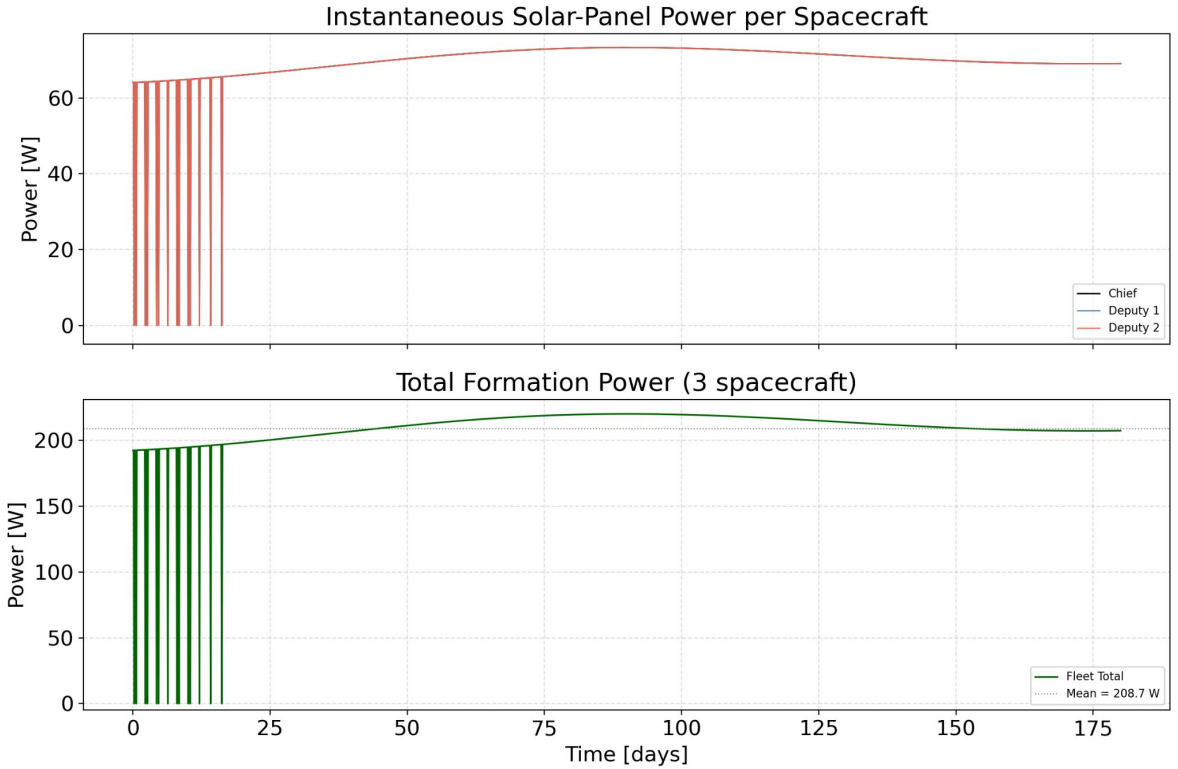
Methodology

- Review idealized force model deficiencies
- Build on premises of idealized force model
- List complexity features added in
  - Higher resolution gravity field
  - Solar radiation pressure
  - nth body effects from Sun and Moon
  - Atmospheric drag (discuss why or why not it's accounted for)
- Complexity coded into already-built idealized simulations
- Implemented via Python's Orekit Wrapper

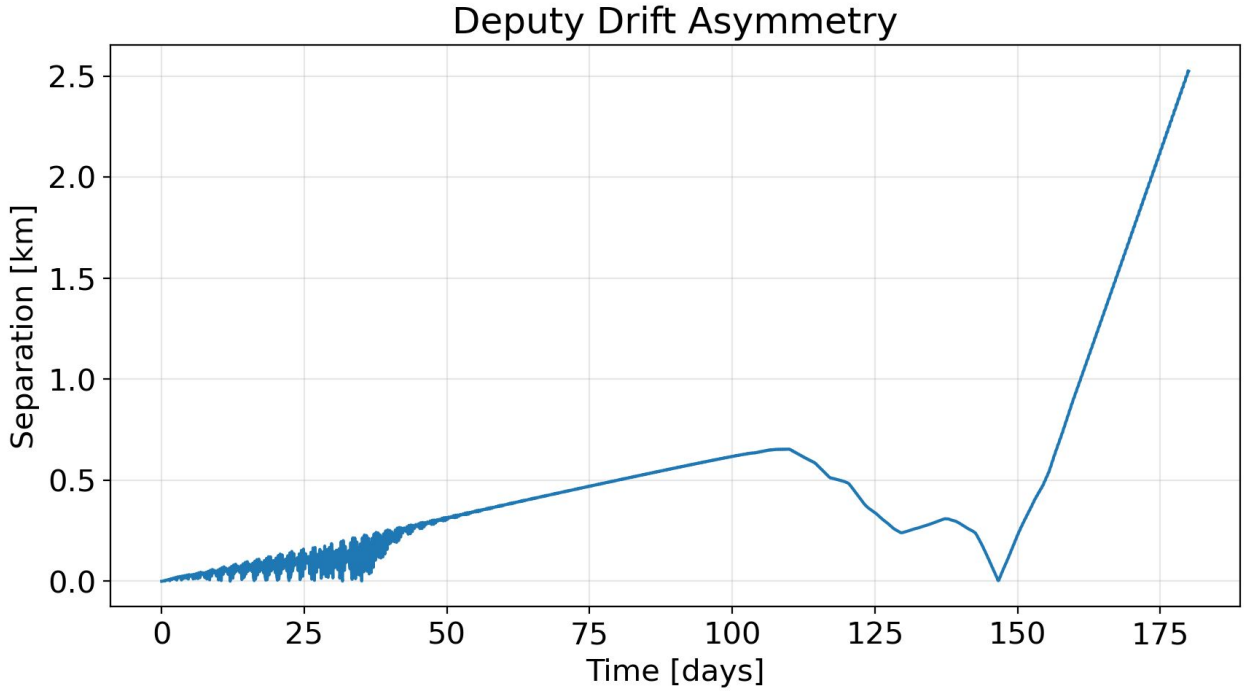
Full Force Model Results

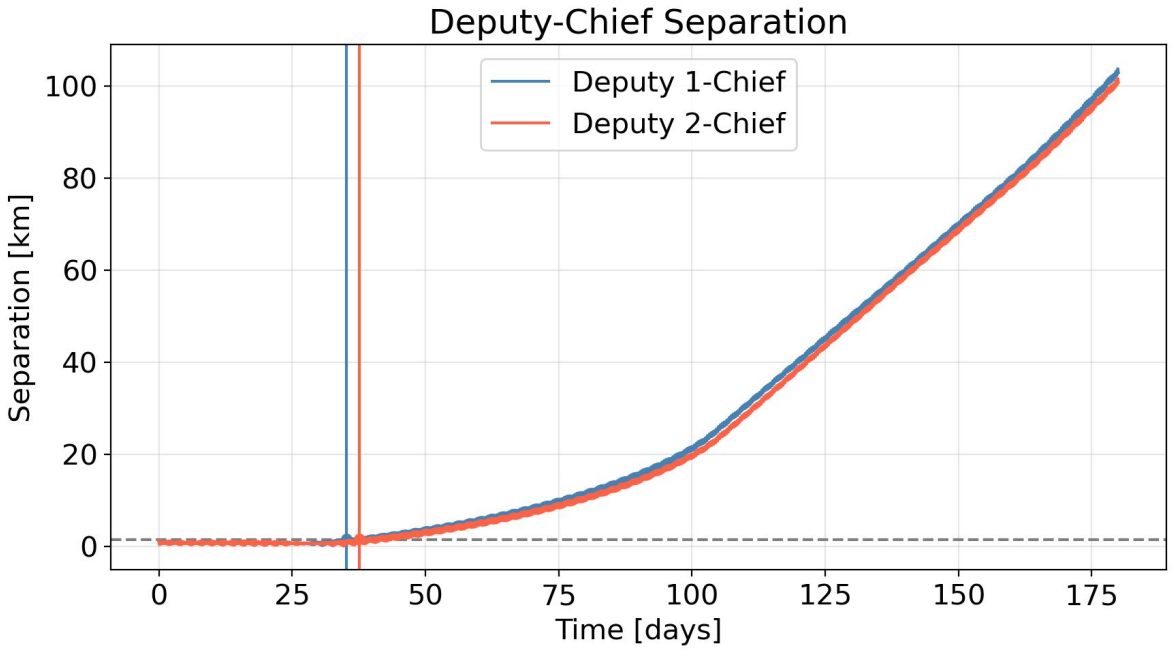
Design A - Trajectory and Solar Power Input



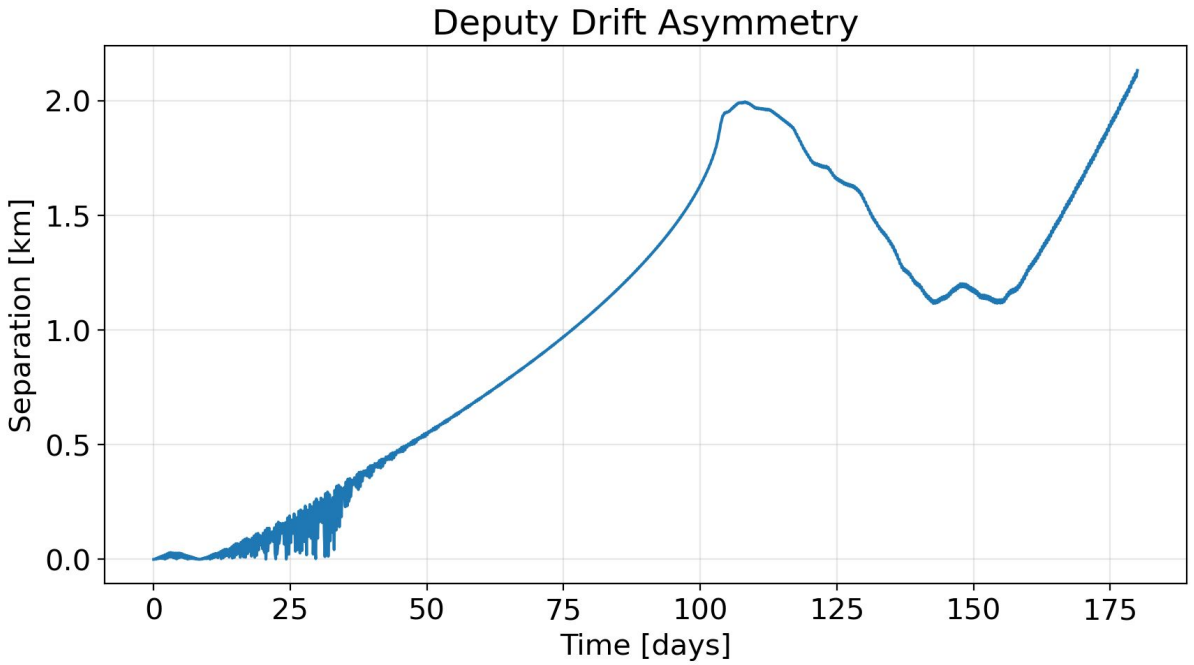


Design A - Asymmetries and Separation

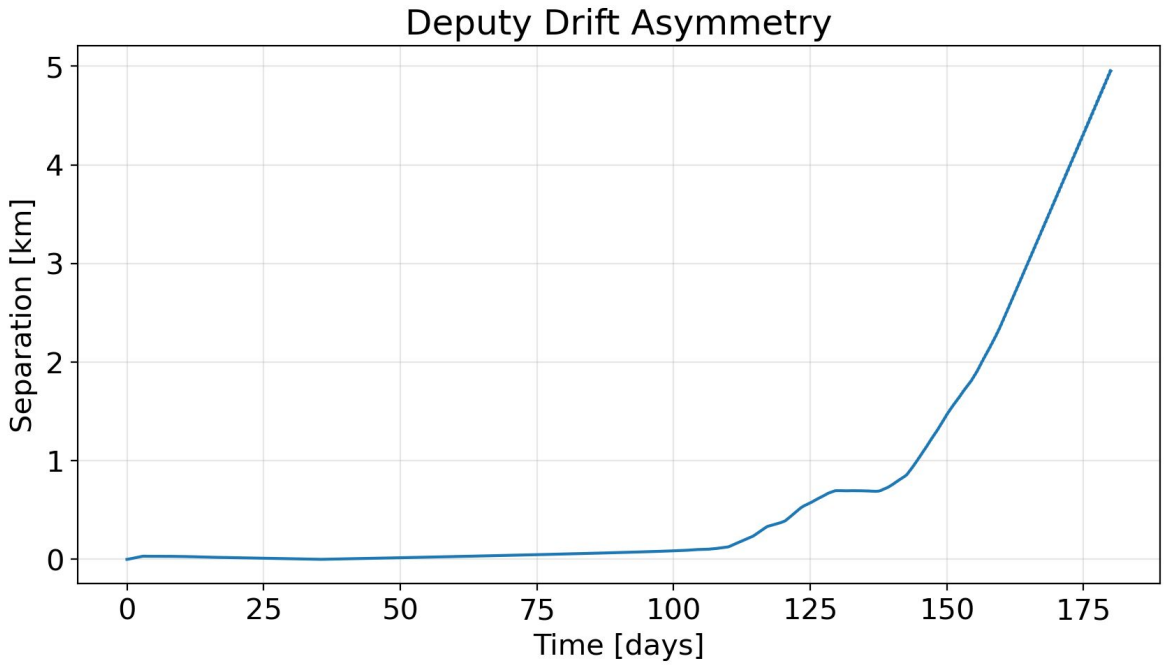




Design A – Helix vs String of Pearls

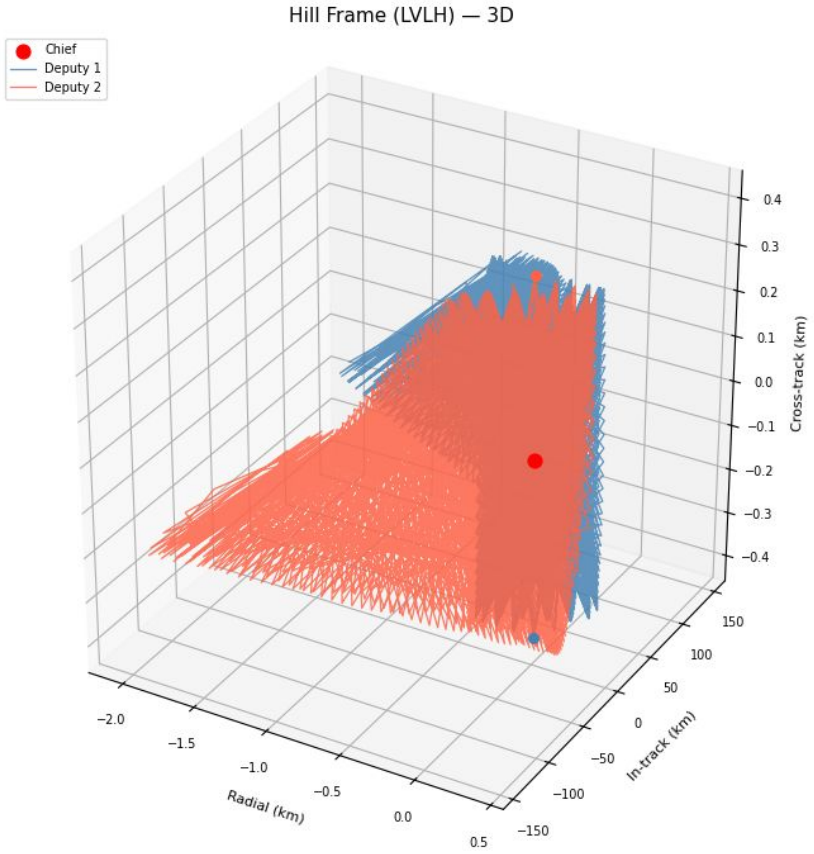


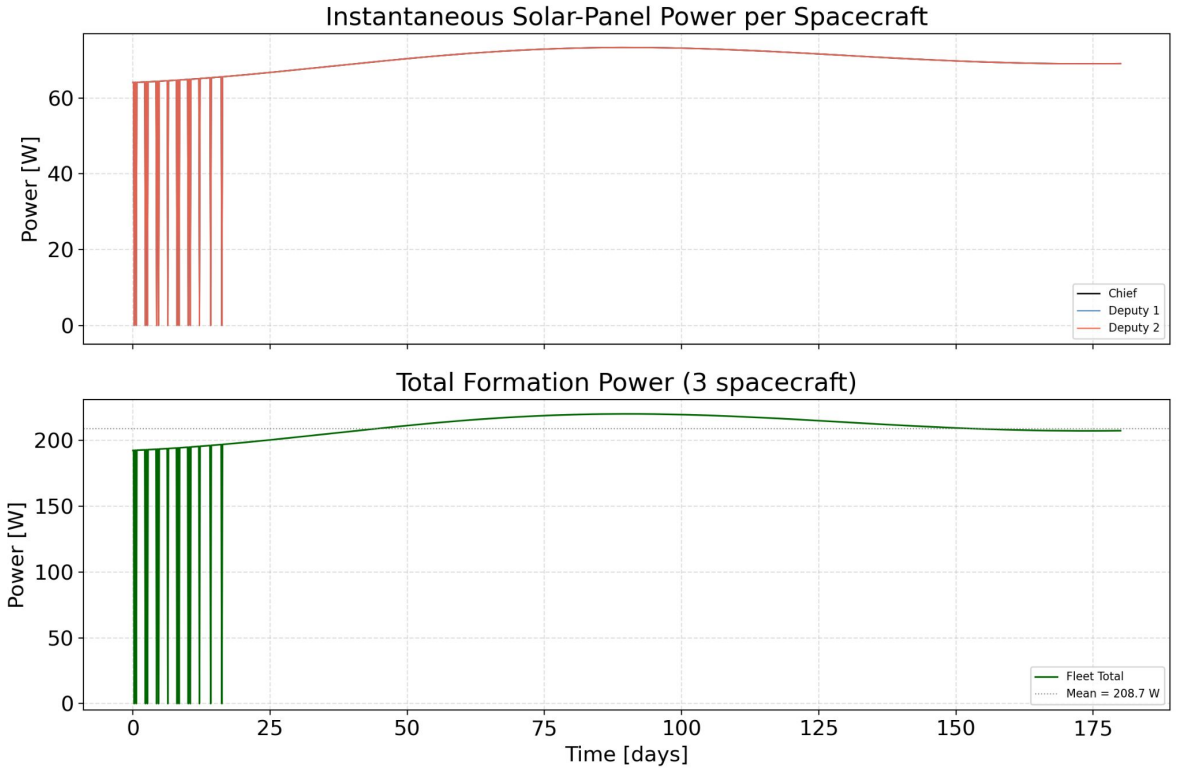
Helix Formation



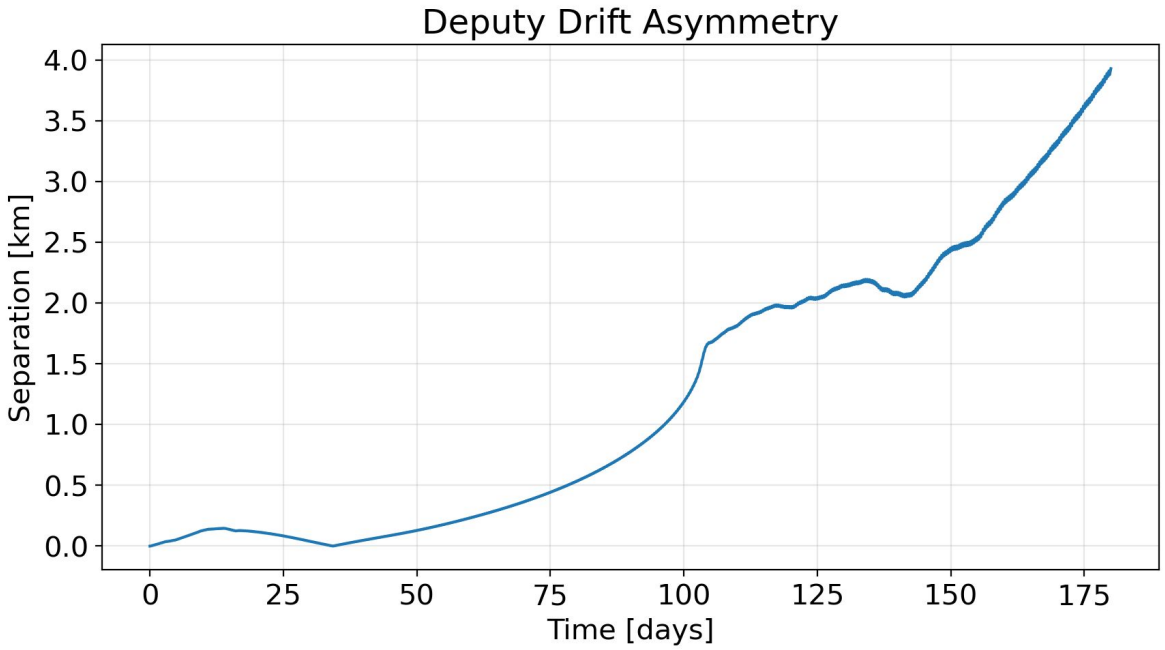
String of Pearls

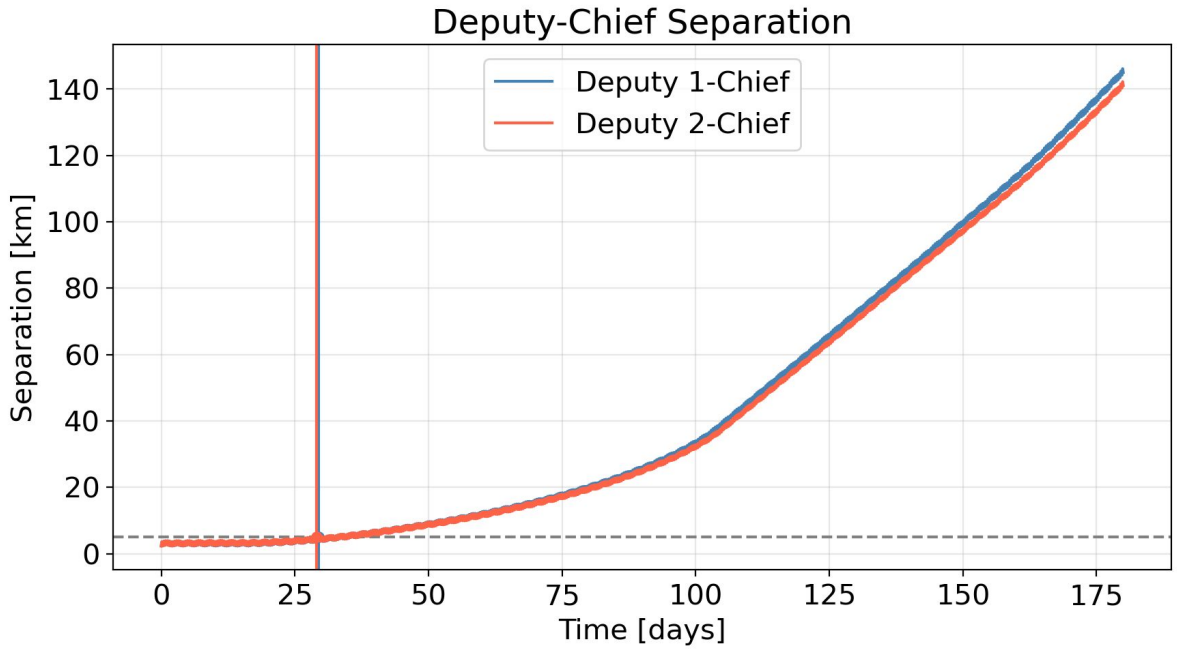
Design B - Trajectory and Solar Power Input





Design B - Separations and Asymmetries





Impulsive Formation Maintenance

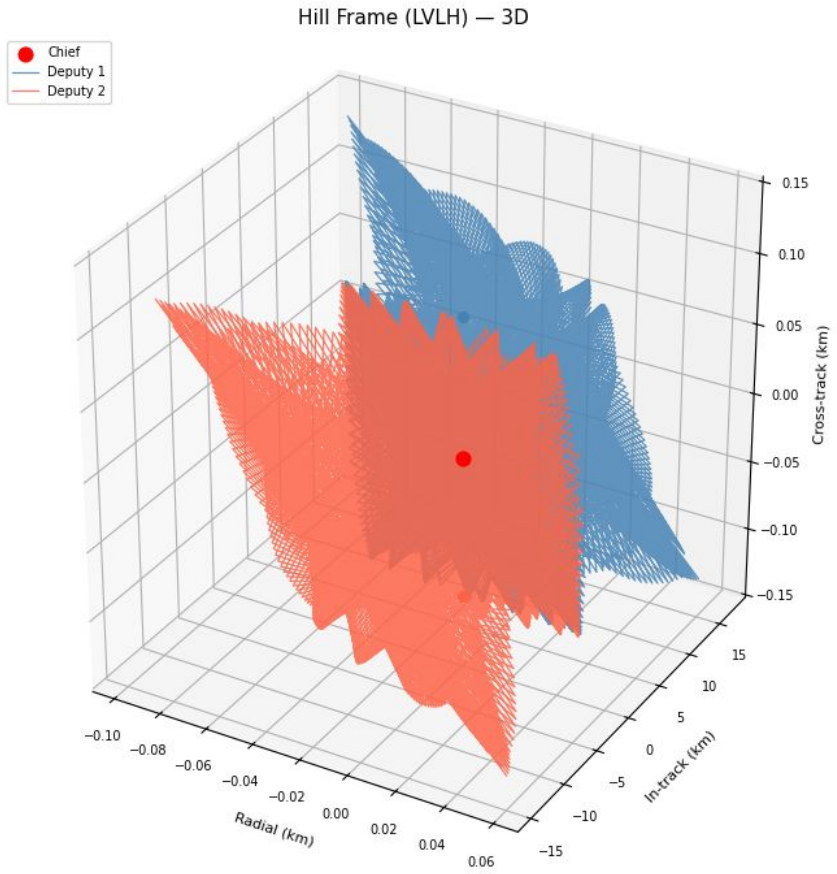
Methodology

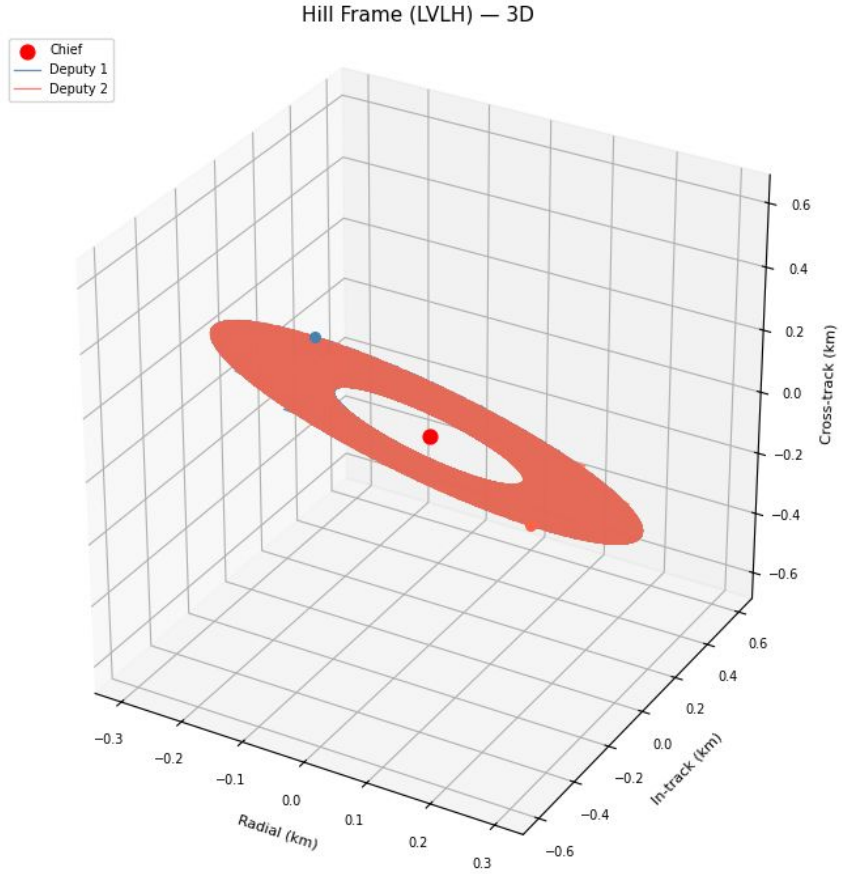
- State-space representation in place of HCW equations
- Design A - pre-computed burns
- Design B - state-triggered burns
- Additionally considering ΔV
- Additional considerations caused by perturbations:
  - Fuel requirements
  - Engine selection possibilities
  - Power generated over course of mission

$$\begin{bmatrix} \boldsymbol{\Phi}_{F,1} & \cdots & \boldsymbol{\Phi}_{F,m-1} & \boldsymbol{\Phi}_{F,m} \end{bmatrix} \begin{pmatrix} \Delta x_1 \\ \Delta x_2 \\ \vdots \\ \Delta x_m \end{pmatrix} = \boldsymbol{b}_0 = (\boldsymbol{x}_F - \boldsymbol{\Phi}_{F,0}\boldsymbol{x}_0) \tag{A1}$$

$$\delta\alpha(t_{k+1}) = \delta\alpha(t_k^-) + B(t_k) \cdot \Delta v(t_k)$$

Design A - Planned Burns

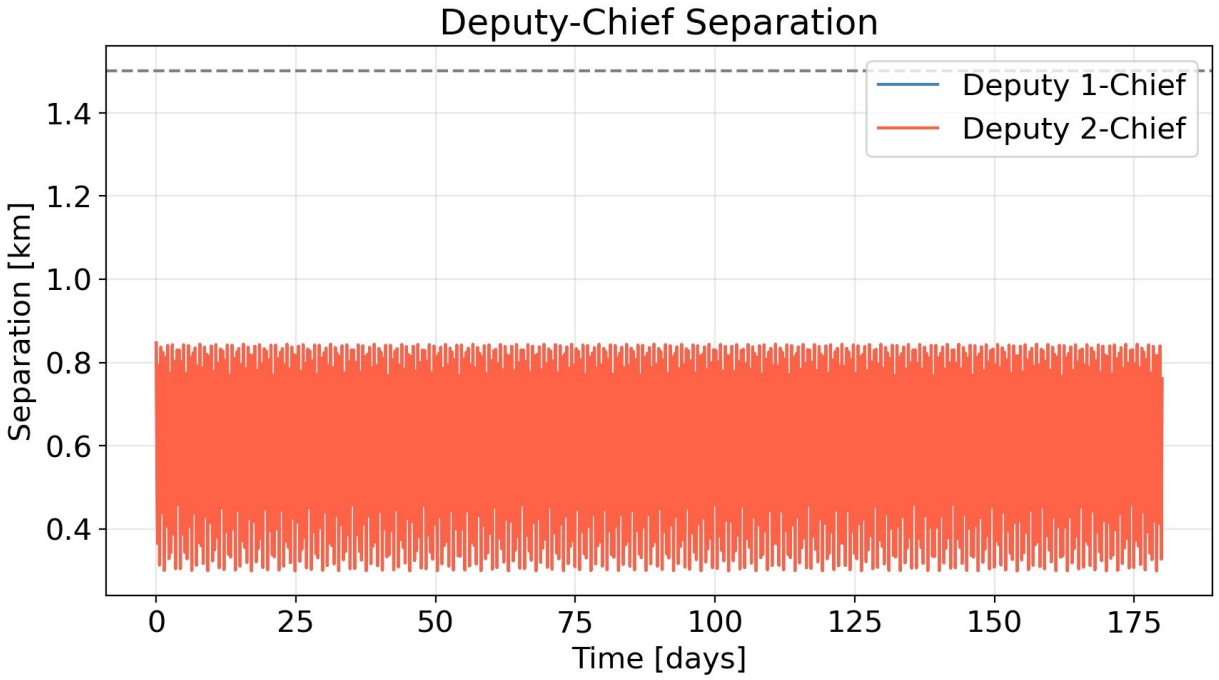




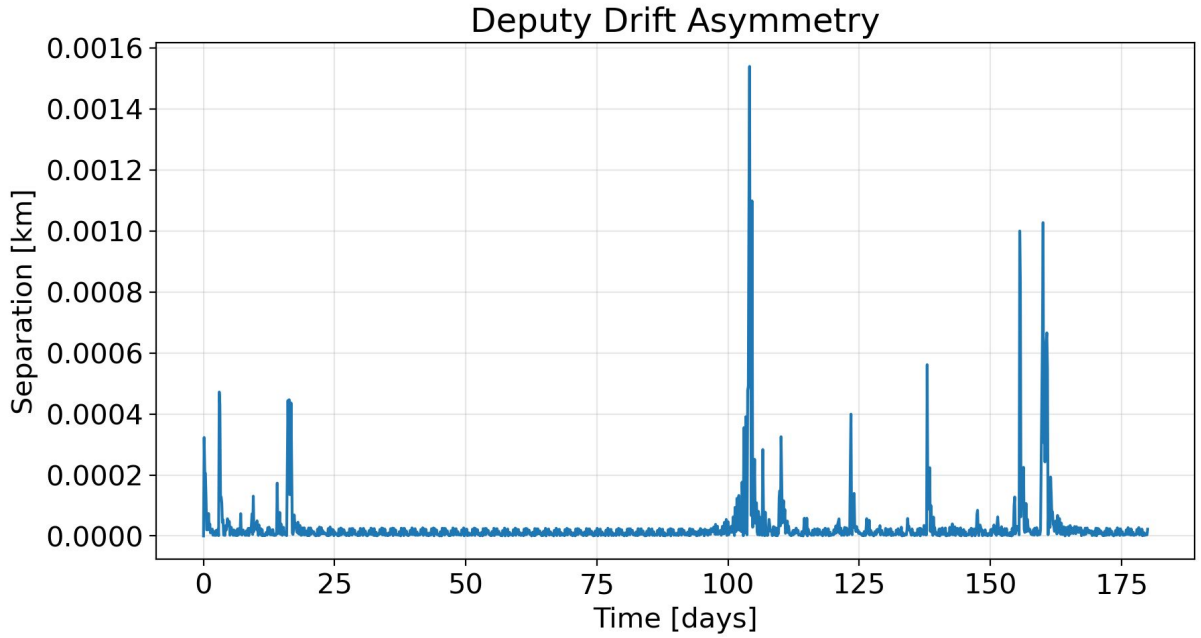
Trajectory of deputies in the Hill Frame

(Before and After Impulsive Maneuver Scheme Implemented)

Design A - Separations and Asymmetries

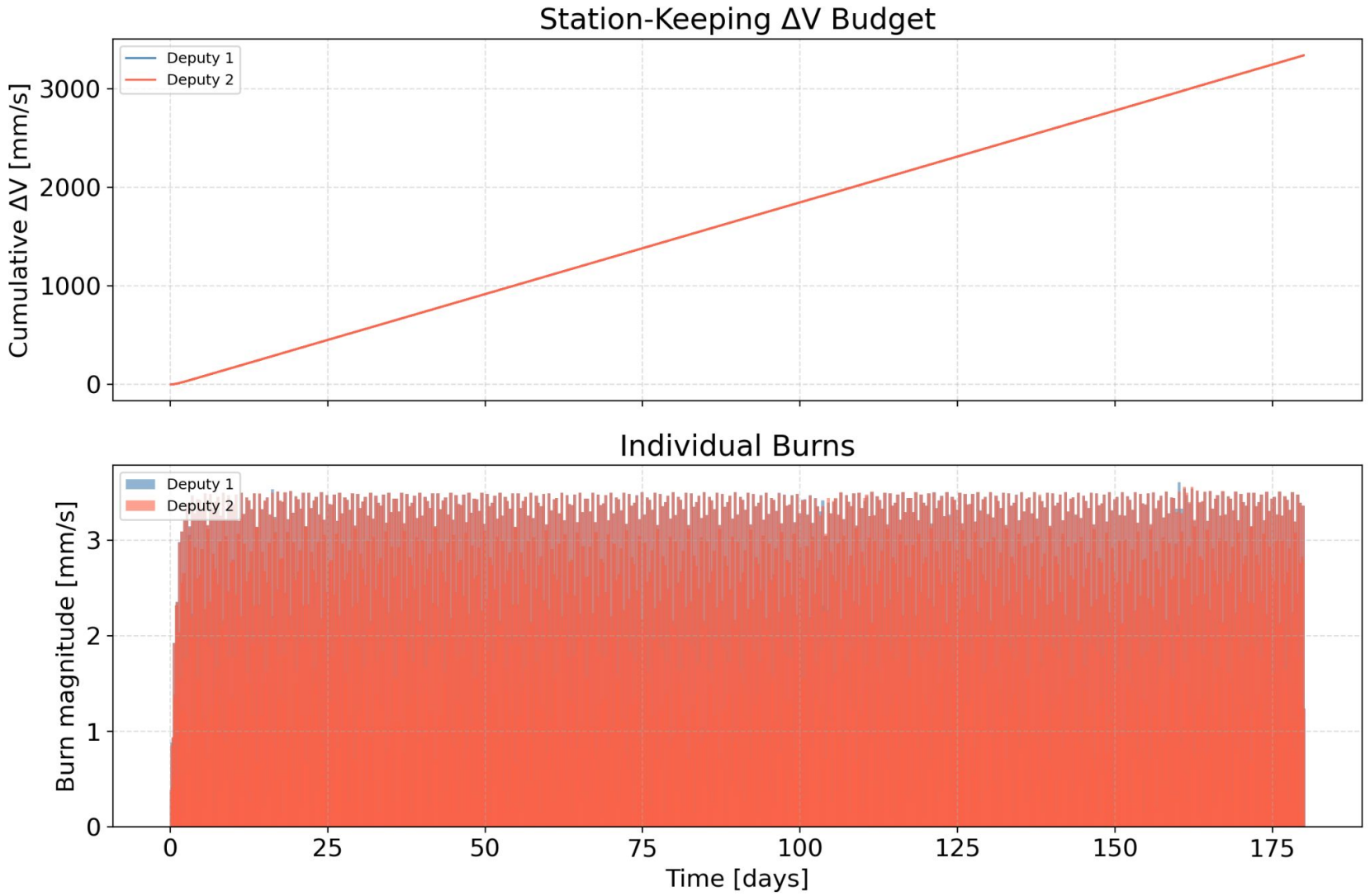


Magnitude of separation from the chief

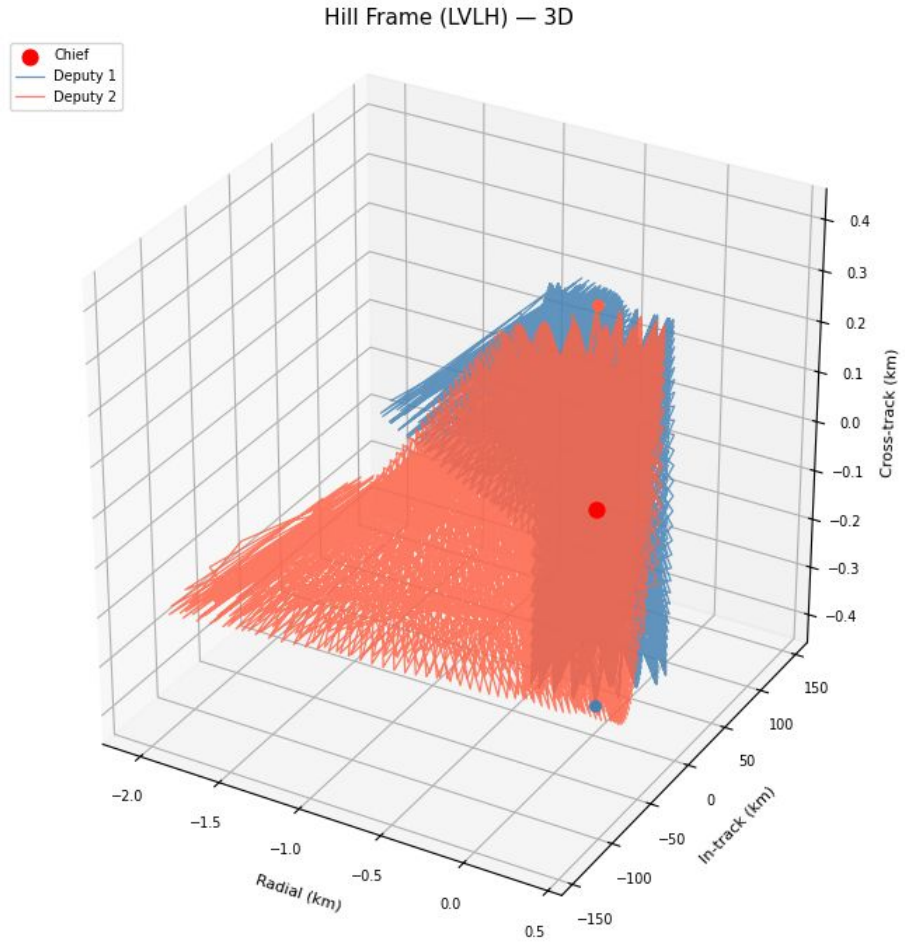


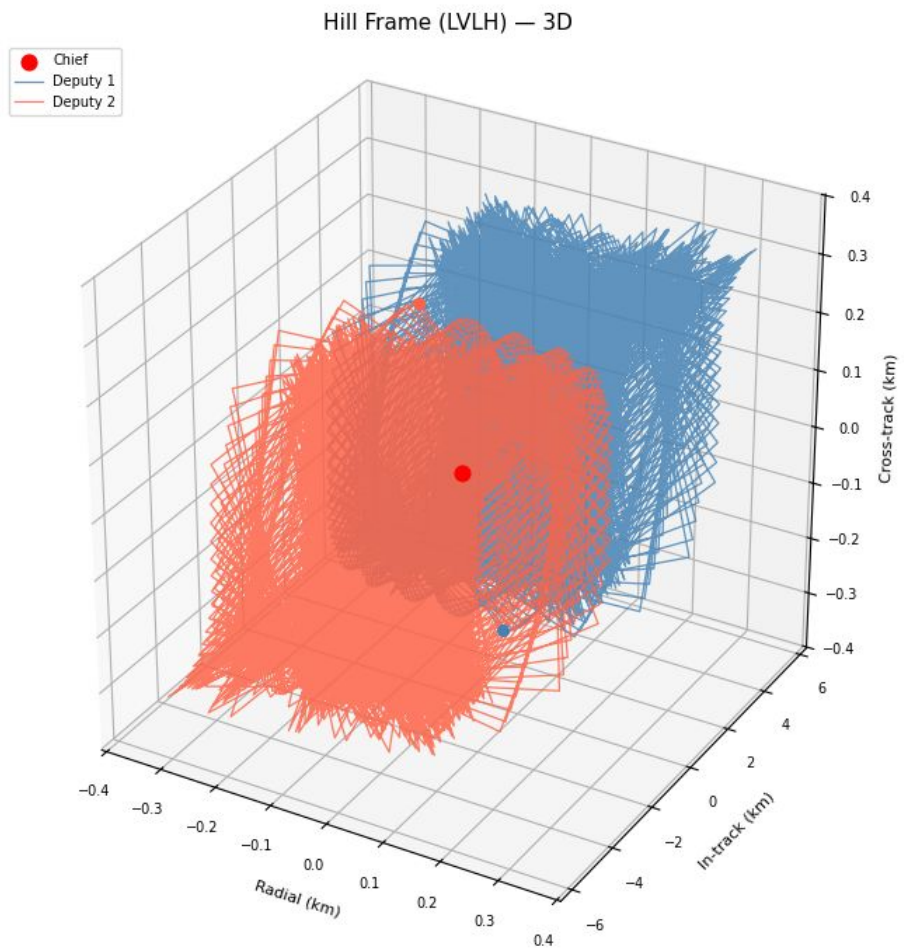
Asymmetry of deputy orbits

Design A – Burns and Cumulative ΔV Budget



Design B - State-Triggered Burns

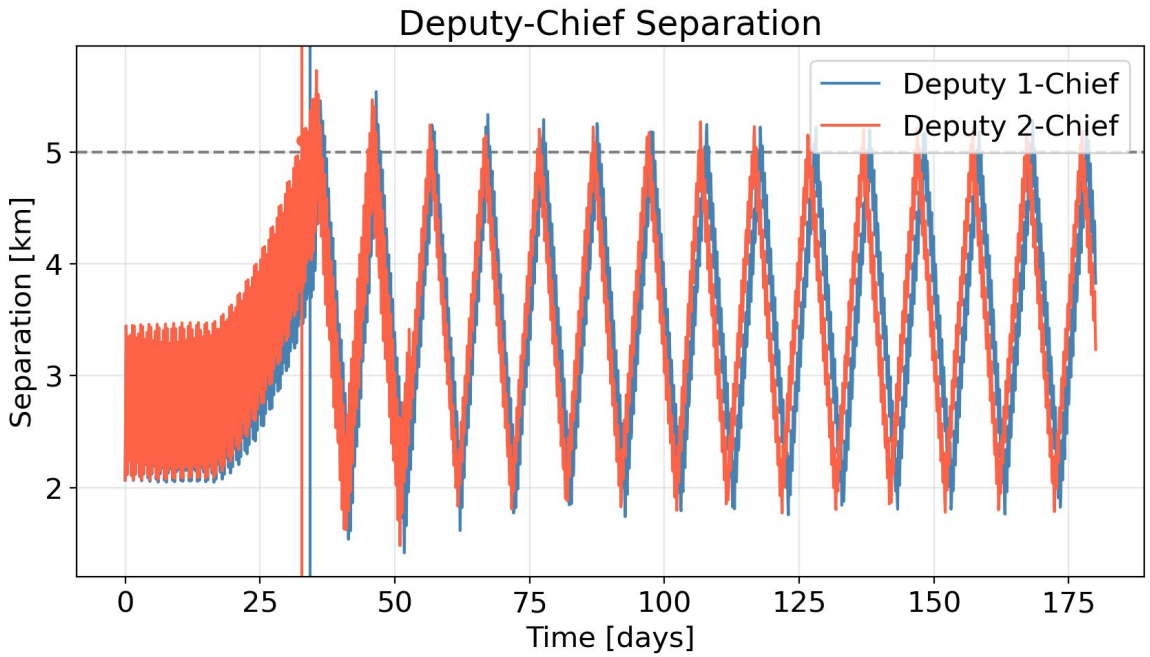




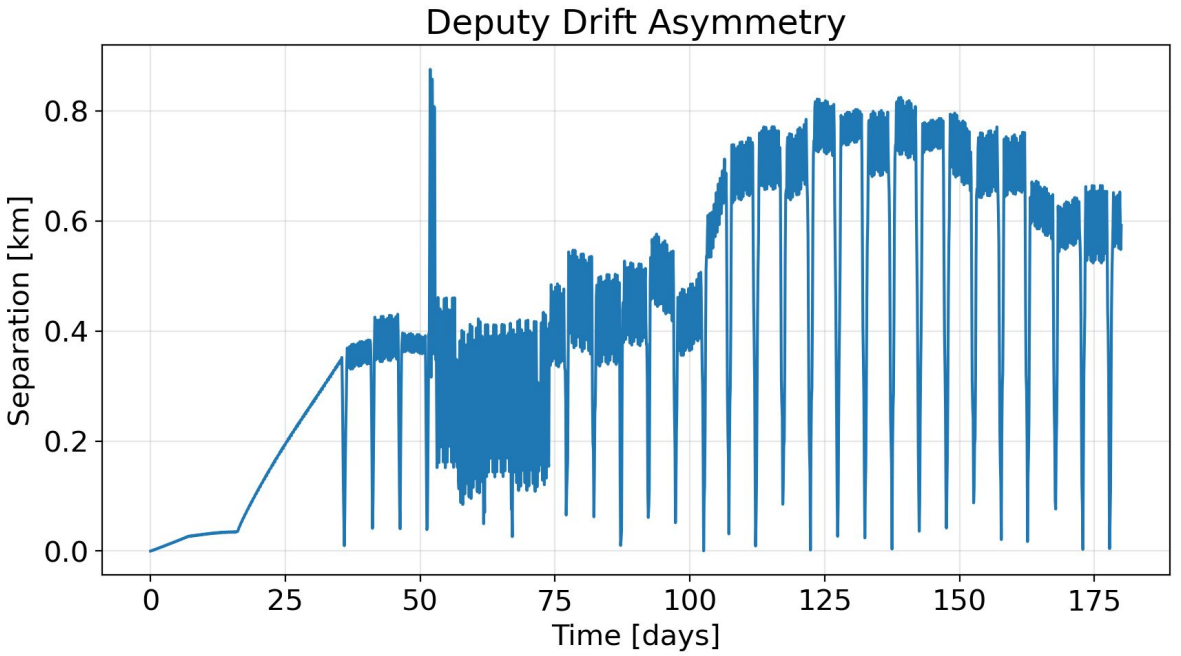
Trajectory of deputies in the Hill Frame

(Before and After Impulsive Maneuvers)

Design B - Separations and Asymmetries

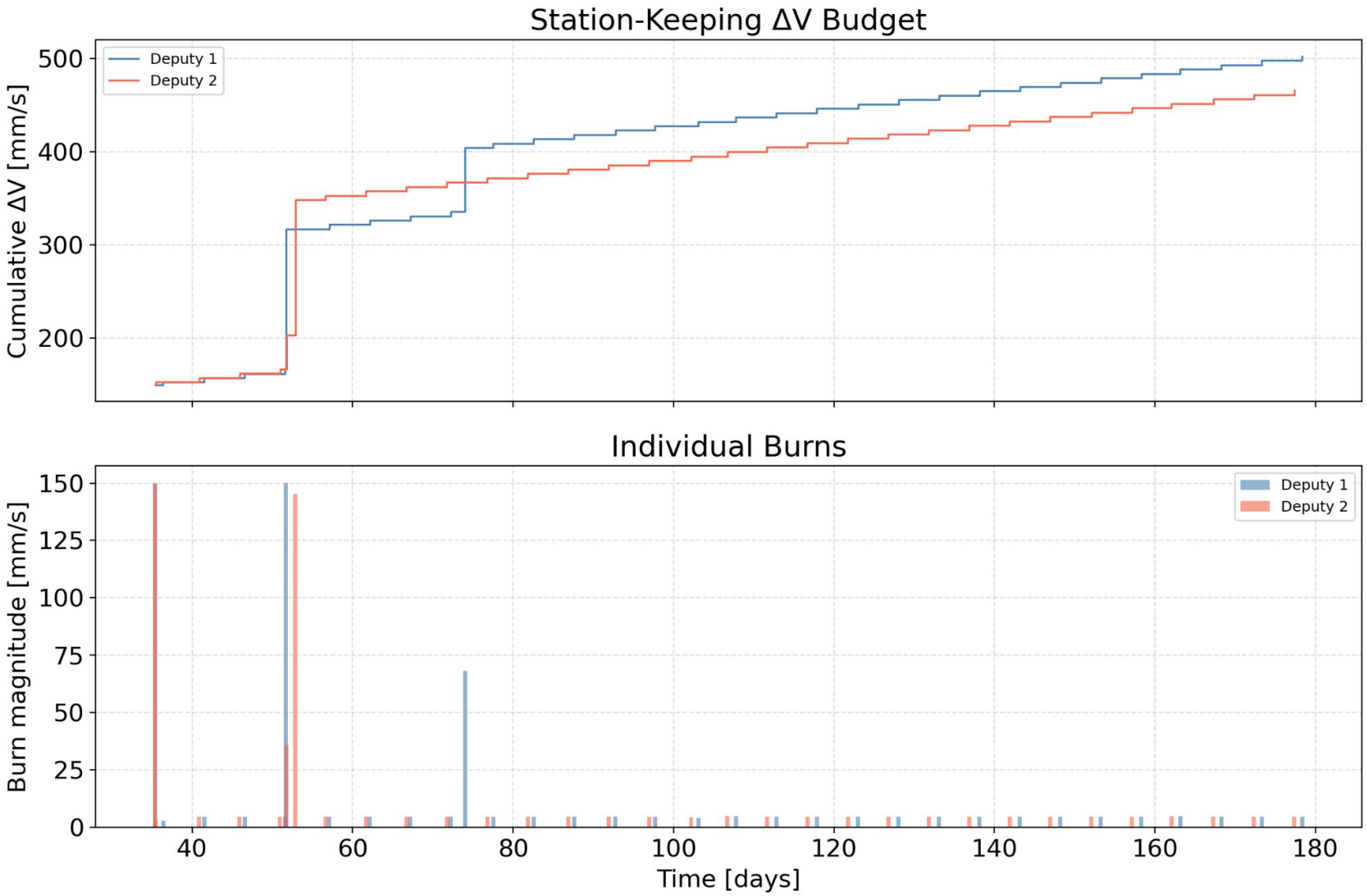


Magnitude of separation from the chief



Asymmetry of deputy orbits

Design B - Burns and Cumulative ΔV Budget



Discussion of Results

Orbital Stability Analysis

- Under J2, our design minimized secular drift, and the relative orbits of the deputies stayed closed and well-behaved.
- However, updating the model to include solar radiation pressure, atmospheric drag, and third body effects from the Sun and Moon meant that there were perturbations that just J2 invariance could not cancel and therefore we saw significant secular drift.
- We see this more in Design A than in Design B as the smaller bounding box for Design A meant that the perturbations were larger with relation to the formation size, so the geometries distorted faster.

Orbital Stability Analysis

- The purpose of the impulsive maneuvers are to cancel out the perturbations to keep the formation geometry as close to the original as possible for the longest amount of time.
- However, the two designs required different types of maneuvers to accomplish that same objective.
  - Design B requires few, well-spaced impulses to maintain formation
  - Design A requires many more, but smaller impulses, presumably because the tighter the formation, the less drift that can occur before the system requires correction.

Design Feasibility

Design A

- Control scheme left unimpeded:
  - Cold gas propulsion
  - Hall effect thruster

Design B

- In-track separation leads to minimal station keeping:
  - Hot gas propulsion
  - Array of electrospray thrusters

Summary and Future Work

- Statements on viability
  - We modeled a Chief - 2 Deputy System with two designs, initially with J2 Perturbations, and then with other natural perturbations
  - We found that while we could minimize secular drift from J2, we could not do so with all of the perturbations, inducing the need for impulsive maneuvers
  - We then designed the maneuvers that would be required to maintain each type of formation.
- Interesting results
  - Helix better than string of pearls at closer proximity, but string of pearls proved better in larger bounding boxes
- Further work
  - The station-keeping for design A, as stated before, requires many impulses at minute intervals. However, it could be possible to model this as a continuous low-thrust situation that could likely yield the same results.
  - Orbital Data Centers probably require more time in orbit than 6 months, so that time considerations should be looked at when considering orbit designs and impulsive maneuvers as well.

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